We sometimes face constrained optimization problems:

$$(q) \quad \min_x f(x) \quad \text{s.t.} \quad g_i(x) \le 0 \quad \text{for all } i = 1, \dots, m$$

→ this form is known as the standard form, all other constrained optimization problem can be formulated as it.

For example, $\max_x f(x)$ can be transformed into it by considering equivalently $\min_x -f(x)$

equation constraint can be transformed $g(x) = 0$ → $g(x) \le 0$ and $-g(x) \le 0$

A simple (but might not be practical) way to solve (q) is to convert it into a unconstrained problem by considering the penalty method.

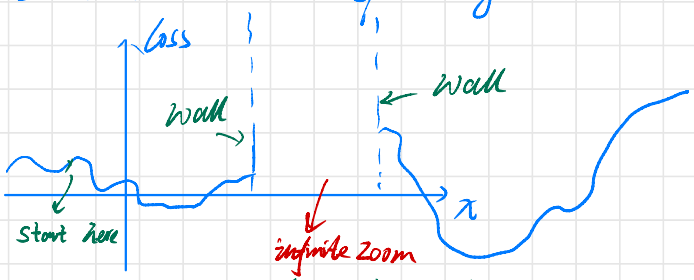
$$\min_x \left(f(x) + \sum_{i=1}^{n} I(g(x_i)) \right)$$ ← add penalty here

Where $I(x) = \begin{cases} 0 & \text{if } x \le 0 \\ +\infty & \text{otherwise} \end{cases}$

↳ or a super large value

Why the naive penalty method often doesn't work well?

(It could make the loss space very bad)



if we start left of it, never be able to cross it

Lagrange multiplier method:

Let us consider the lagrange form:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T g(x), \text{ where } \lambda = (\lambda_1, \dots, \lambda_n)^T$$
$$g(x) = (g_1(x), \dots, g_n(x))^T$$

Then, we have the original problem is equivalent to the following problem:

$$\min_x \max_\lambda \mathcal{L}(x, \lambda)$$
$$s.t. \quad \lambda \geq 0$$

note that for any $\varphi(x, y)$, we have:

$$\min_x \varphi(x, y) \leq \varphi(x, y) \leq \max_y \varphi(x, y)$$

$$\Rightarrow \max_y \min_x \varphi(x, y) \leq \min_x \max_y \varphi(x, y)$$

Hence, we have the solution for the following Lagrange Dual problem:

$$(d) \qquad \max_\lambda \min_x \mathcal{L}(x, \lambda)$$
$$s.t. \quad \lambda \geq 0$$

will be smaller or equal to the original problem (p).

Let us compare the original problem (q) and dual problem (d).

Denote $q(\lambda) = \min_x \mathcal{L}(x, \lambda)$ (this unconstrained problem for x is known as inner problem)

Say we can solve it (often possible in analytics form)

We can see the dual problem (d) is much easier than the original problem (q).

(1) The constraints goes from

$g_i(x) \le 0 \longrightarrow \lambda_i \ge 0$

($g_i(x) \le 0$: Could be very complex)

($\lambda_i \ge 0$: this is known as "box constraints" the easiest constraints)

(2) $\min_x f(x)$ Could be a non-convex problem

$\max_\lambda q(\lambda)$ is a convex problem for λ.

(3) Dimensionality reduction.

Say $x \in \mathbb{R}^{10,000}$ and we have 5 constraints, then $\min_x f(x)$ could be very hard

By considering $\max_\lambda q(\lambda)$, we reduce the dimension into 5.

In short, the dual problem could be dramastically simpler than the original problem.

However, keep in mind that the results for dual problem (d) Could be smaller than the original problem (q)

The difference between the results for original and dual problem is

duality gap: $G = p^* - d^*$ (p^*: results for original problem; d^*: results for dual problem)

if $G = 0$ ⇝ Strong duality

$G > 0$ ⇝ Weak duality (this means we have a lower bound for the original problem, which implies the x found in the dual problem is not in the feasible region of the original problem)

↓ this could happen because in the inner problem

$\min_x \left(f(x) + \lambda^T g(x) \right)$

↓ when violate $g_i(x) < 0$, get penalty $\lambda_i g_i(x)$. however if $f(x_i)$ is very deep (i.e. very small) this violation might worth it...

When will Strong Duality hold?

A necessary but not sufficient condition is known as

Slater's Condition:

1. Convex: $f(x)$ and $g_i(x)$ are convex

2. Feasibility: there is at least one x such that $g_i(x) < 0$ for all $i = 1, \dots n$

↓ strictly smaller

While the Lagrange method gives us a often simpler dual problem, it does not solve the problem directly.

1. if we have only equation constraints

$$\min_x f(x)$$

s.t. $g_i(x) = 0$, $i = 1, \dots m$

$g_i(x) = 0$ → $g_i(x) \geq 0$, $g_i(x) \leq 0$

⇒ Lagrange form:

$$\mathcal{L}(x, \lambda, \lambda') = f(x) - \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{i=1}^{n} \lambda_i' g_i(x)$$

s.t. $\lambda_i \geq 0$, $\lambda_i' \geq 0$

$$\Leftrightarrow \mathcal{L}(x, u) = f(x) - \sum_{i=1}^{m} u_i g_i(x)$$

if we have only equation constraints, we get unconstrained optimization problem

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \rightarrow n \text{ equations}$$

$$\frac{\partial \mathcal{L}}{\partial u_i} = g_i(x) = 0 \rightarrow m \text{ equations}$$

→ solve them for the optimal x.

2. if we have inequality constraints

$$\min f(x)$$

s.t. $g_i(x) = 0$

$h_i(x) \leq 0$

$$\Rightarrow \mathcal{L}(x, \lambda, u) = f(x) + \sum_{i=1}^{m_1} u_i g_i(x) + \sum_{i=1}^{m_2} \lambda_i h_i(x)$$

s.t. $\lambda_i \geq 0$

1. if $h_i(x) < 0$ (constrained strictly satisfied) then in order to maximize $\mathcal{L}(x, \lambda, u)$ we must have $\lambda_i = 0$.
2. if $h_i(x) = 0$, then we have $\lambda_i \geq 0$

→ we must have $\lambda_i h_i = 0$!

To solve constrained problems with inequality constraints,

We often solve a constrained optimization directly using:

KKT Conditions → gives us a local minimum, if Slater's condition holds, a global minimum

1. Stationarity

$$\nabla_{x_i} f(x) + \lambda^T \nabla_{x_i} g(x) = 0$$

→ Lagrange form

inner constraint problem $\min_x \mathcal{L}(x, \lambda)$

if x minimize it, the gradient must be 0.

2. Primal Feasibility

$$g_i(x) \leq 0 \quad \text{for } i = 1, \dots, n$$

3. Dual Feasibility

$$\lambda_i \geq 0, \quad \text{for } i = 1, \dots n$$

4. Complementary Slackness

$$\lambda_i g_i(x) = 0$$

→ if $g_i(x) < 0$, then $\lambda_i = 0$

if however $g_i(x) = 0$, then $\lambda_i \geq 0$

recall that the original problem is equivalent to

$$\min_x \max_{\lambda \geq 0} \mathcal{L}(x, \lambda)$$

$$= \min_x \max_{\lambda \geq 0} \left(f(x) + \lambda^T g(x) \right)$$

1. Stationarity

if local minimum, then the gradient must be 0 (for every x_i)

2. Complementary Slackness

if $g_i(x) < 0$, then $\lambda_i = 0$

because for fixed x, λ maximize $\mathcal{L}(x, \lambda)$.

Steps for using KKT Conditions:

(Sufficient condition)

KKT conditions are a checklist whether x^*, λ^* we have in hand result valid local (or global if Slater's condition holds) results. But it can also be used to find the local minimum. The idea is to

Construct x^*, λ^* to satisfy KKT conditions.

Steps:

1. Guess which points are on the boundaries (following Complementary Slackness)

2. Solve x and λ (λ: k_2) from the stationarity and complementary slackness

   (x: say k_1; stationarity: k_1 equations; complementary slackness: k_2 equations)

3. Check feasibility. if pass, find a local minimum,

   if not, start from step 1 again.

   note that different guess in step 1 could give us different local minimum. In general, there are total 2^{k_2} different combinations.

   (k_2: number of constraints)

A simple example of constrained problem:

$$\min_{\hat{p}_1,\dots,\hat{p}_K} \sum_{i=1}^{K} -p_i \log(\hat{p}_i)$$

s.t. $\sum_{i=1}^{K} \hat{p}_i = 1$

$\hat{p}_i \geq 0$, for all $i = 1,\dots,K$

$$\mathcal{L}(\hat{p}, \mu, \lambda) = -\sum_{i=1}^{K} p_i \log(\hat{p}_i) - \mu\left(\sum_{i=1}^{K} \hat{p}_i - 1\right) - \sum_{i=1}^{K} \lambda_i \hat{p}_i$$

$$= -\sum_{i=1}^{K} p_i \log(\hat{p}_i) - \mu\left(\sum_{i=1}^{K} \hat{p}_i - 1\right)$$

Complementary slackness condition

either $\lambda_i = 0$

or $\hat{p}_i = 0$ → this would lead the results to ∞

$$\frac{\partial \mathcal{L}}{\partial \hat{p}_i} = \frac{p_i}{\hat{p}_i} - \mu = 0$$

$$\Rightarrow \hat{p}_i = \frac{p_i}{\mu}$$

Since $\sum_{i=1}^{K} \hat{p}_i = 1 \Rightarrow \sum_{i=1}^{K} \frac{p_i}{\mu} = 1 \Rightarrow \mu = \sum_{i=1}^{K} p_i$

$$\Rightarrow \hat{p}_i = \frac{p_i}{\sum_{i=1}^{K} p_i}$$